

Task 3: Data Cleaning Report

Objective

The objective of this task was to clean and preprocess the Airbnb NYC 2019 dataset. Data cleaning is an important step in the data analysis process because it improves data quality, consistency, and reliability for future analysis.

Dataset Description

The dataset used in this project is the Airbnb NYC 2019 dataset, which contains information about Airbnb listings in New York City. The dataset includes details such as listing names, host information, room types, prices, reviews, and availability.

Data Cleaning Process

1. Dataset Loading and Exploration

The dataset was loaded into Python using the Pandas library. Initial exploration was performed using functions such as `head()`, `shape()`, and `info()` to understand the structure of the dataset.

2. Missing Value Analysis

Missing values were identified in the following columns:

- name
- host_name
- last_review
- reviews_per_month

These missing values were analyzed before applying suitable treatment methods.

3. Missing Value Handling

Missing values in text columns were replaced with placeholder values such as "Unknown", while numerical missing values were filled using the median value of the respective column.

4. Duplicate Record Analysis

The dataset was checked for duplicate records using Pandas functions. Duplicate entries were identified and removed to maintain data consistency.

5. Data Standardization

Text-based columns were standardized to ensure consistent formatting throughout the dataset. This improved readability and data quality.

6. Outlier Detection and Removal

The price column was analyzed using boxplots to identify extreme values. The Interquartile Range (IQR) method was applied to detect and remove outliers that could negatively affect statistical analysis.

7. Date Conversion

The `last_review` column was converted into datetime format to ensure proper date handling and consistency.

8. Exporting the Cleaned Dataset

After completing all cleaning operations, the final cleaned dataset was exported as a CSV file for future analysis and visualization.

Tools and Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook

Results

The dataset was successfully cleaned by:

- Handling missing values
- Removing duplicate records
- Standardizing text data
- Detecting and removing outliers
- Converting date columns into appropriate formats

These preprocessing steps improved the overall quality and reliability of the dataset.

Conclusion

The Airbnb NYC 2019 dataset was successfully cleaned and prepared for further analysis. Proper data cleaning techniques were applied to handle missing values, duplicate records, inconsistent formatting, and outliers. The final dataset is accurate, consistent, and ready for data analysis and visualization.